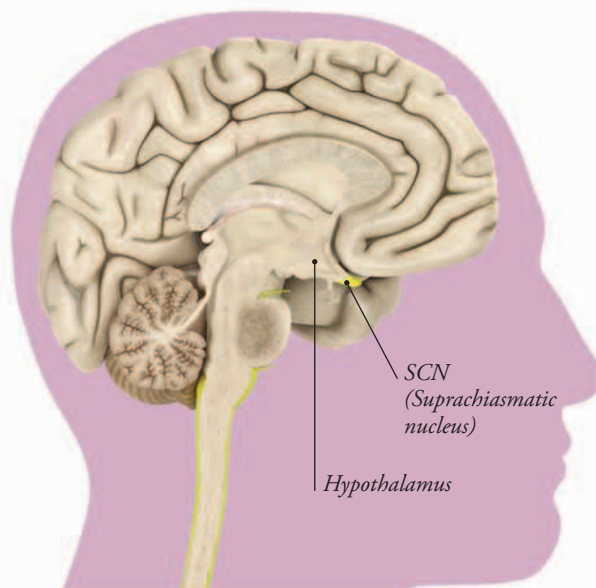


Sleep

Everybody needs sleep. We can't survive without it, and we spend about a third of our lives sleeping. Our eyes close, breathing and heart rate slow down, and we become much less aware of the outside world. No one understands the exact purpose of sleep, but it may give the brain the chance to process the day's information.

WOW!

The world record for going without sleep is 11 whole days, achieved by American student Randy Gardner in 1964.

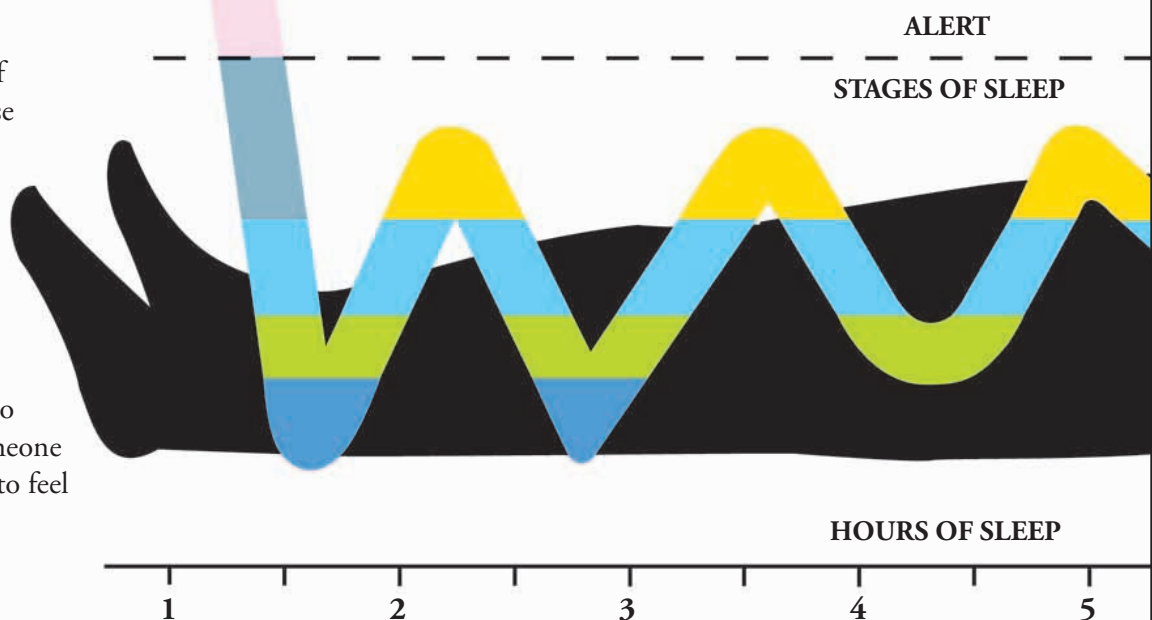


CONTROL CENTER

The daily cycle of sleeping and wakefulness is controlled by a part of the brain called the hypothalamus, which sends signals to many other parts of the brain and body to control their activity. The timing of these signals is set by a part of the hypothalamus called the SCN, which picks up signals from the eyes to determine when day becomes night.

THE SLEEP CYCLE

There are five different stages of sleep, and we pass through these several times a night. Dreams occur during a stage called rapid-eye-movement (REM) sleep, when our brains are highly active. The other four stages of sleep are called non-REM sleep. Stage 4 is the deepest and the most difficult to wake up from. If you wake someone in stage 4 sleep, they are likely to feel groggy and confused.



▲ AWAKE

You are alert and aware of your surroundings when awake. It is the opposite of being asleep.

▲ STAGE 1 SLEEP

This is the lightest stage of the sleep cycle, when you feel drowsy rather than fast asleep.

▲ STAGE 2 SLEEP

As you enter this stage of light sleep, body temperature falls and heart rate slows down.

▲ STAGE 3 SLEEP

In this stage between light and deep sleep, delta brain waves (see right) first appear.